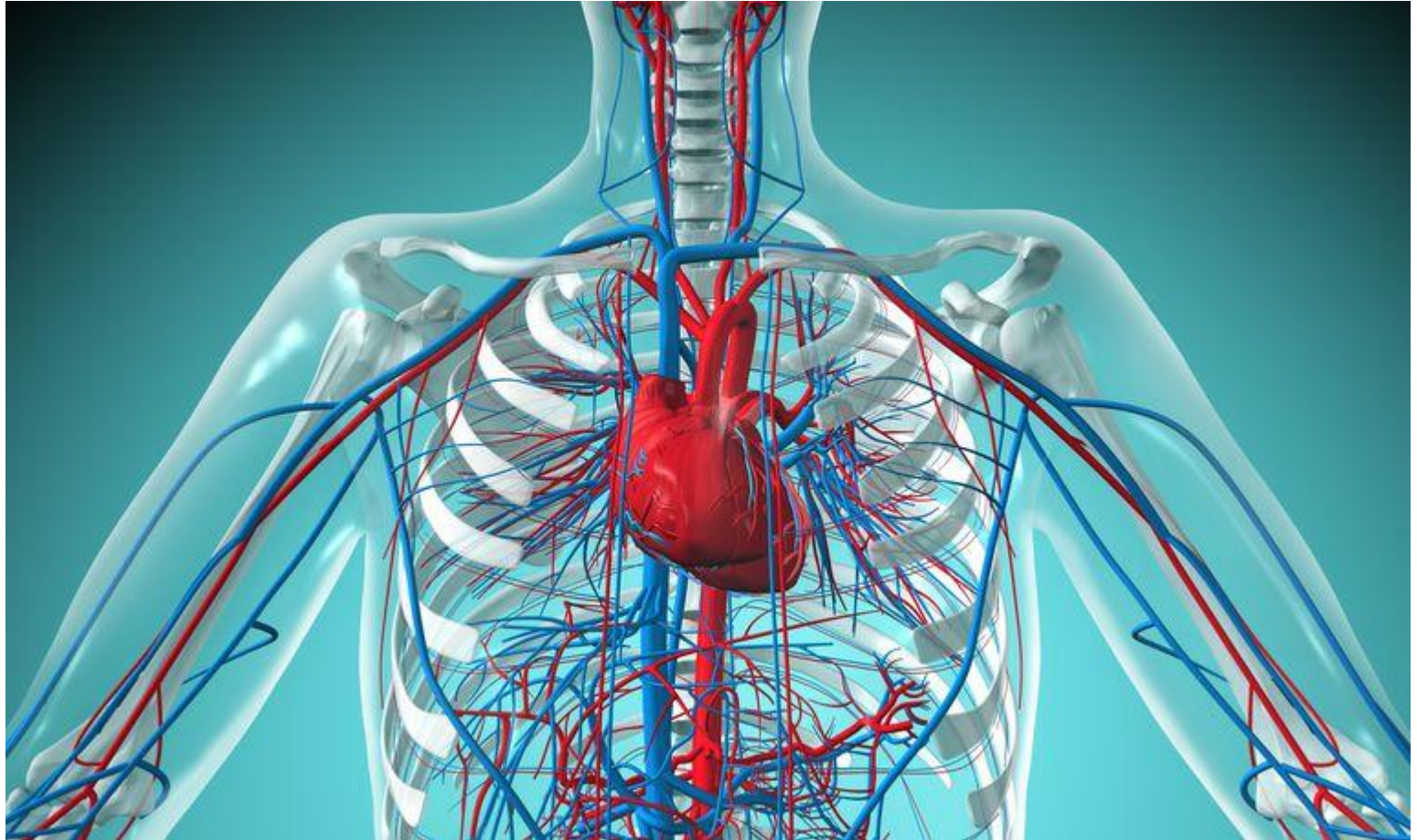


Cardiovascular system



The cardiovascular system includes:

- 1 - the heart, which serves as a pump for the blood.
- 2- the blood vessels, which transport the blood throughout the body

The heart consists of two separate pumps:

1- The right side pump.

- pumps blood to the lungs through the pulmonary circulation so that gas exchange.

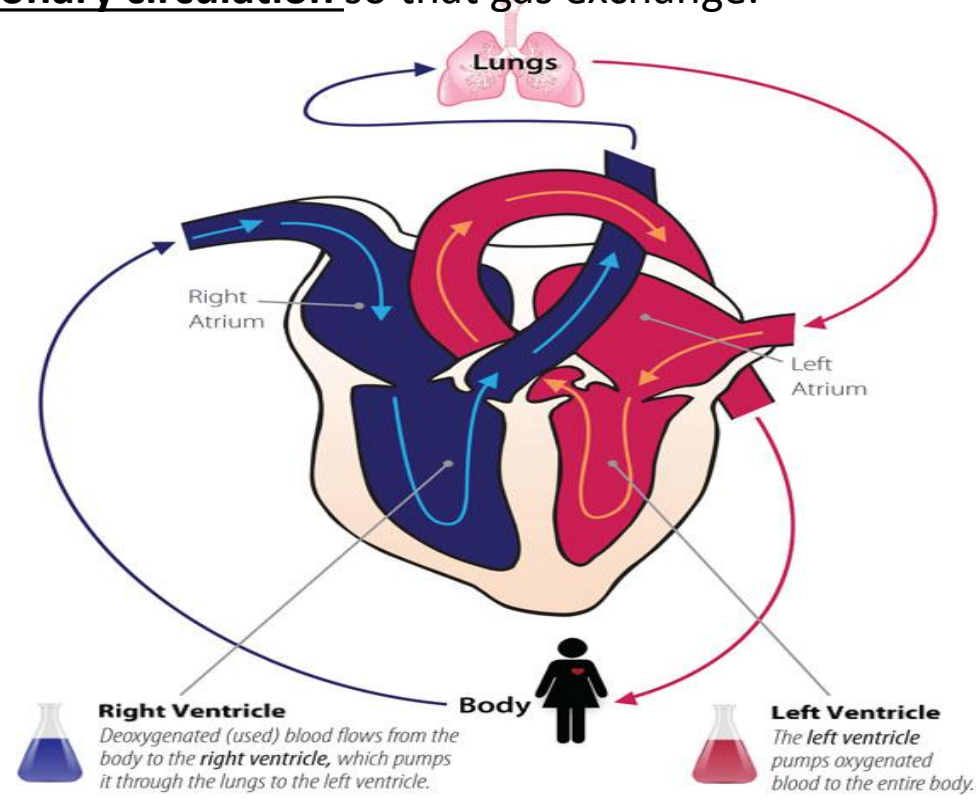
2- The left side pump.

- pumps blood to the rest of the body tissues through the systemic circulation

Notes:

The right side pump... deoxygenated blood

The left side pump.... oxygenated blood



Functional anatomy of the heart

To function mechanically as a pump, the heart must have:

Receiving chambers:

The *atria (singular: atrium)* are the chambers that receive the blood returning to the heart through the veins

Delivery chambers:

The blood then moves to the *ventricles of the* heart, or the delivery chambers

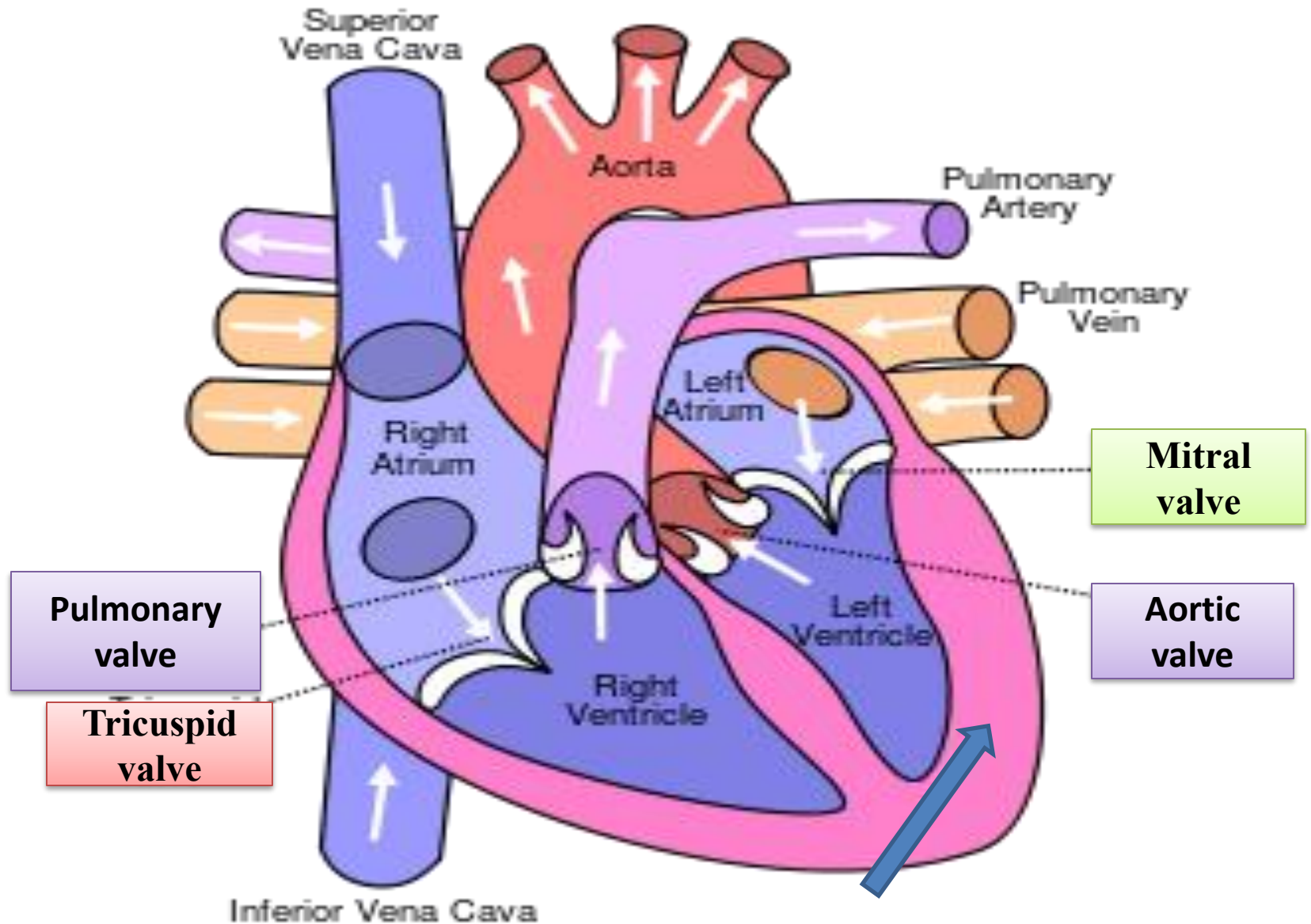
Valves:

Valves ensure the one-way, or forward, flow of the blood.

Atrioventricular valves: Between atria and ventricles (**Tricuspid valve** and *mitral valve*).

Semilunar valves: At the start of the arteries that come from the heart. (*pulmonary valve and aortic valve*)

Types of valves



Heart sound

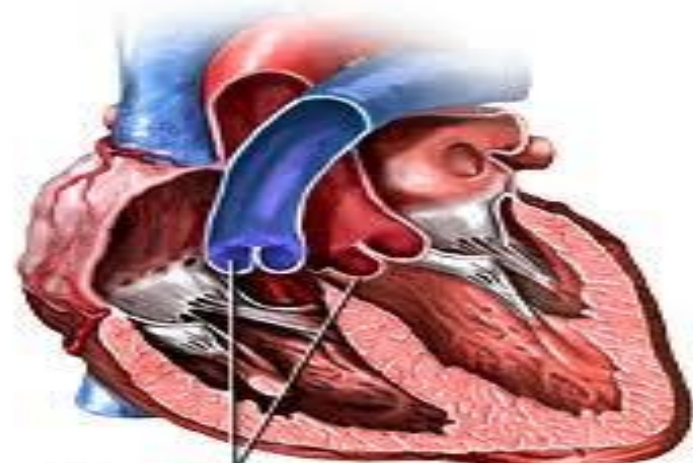
- The closure of the valves causes the “**lub-dub**” associated with the heartbeat.
- The first heart sound, or the “lub,” occurs when the ventricles contract and the AV valves close (closure of the mitral and tricuspid valves).
- The second heart sound, or the “dub,” occurs when the ventricles relax and the semilunar valves close (closure of the *pulmonary valve and aortic valve*).

First heart sound,
“lub”, occurs when
atrioventricular valves
close



Atrioventricular
valves

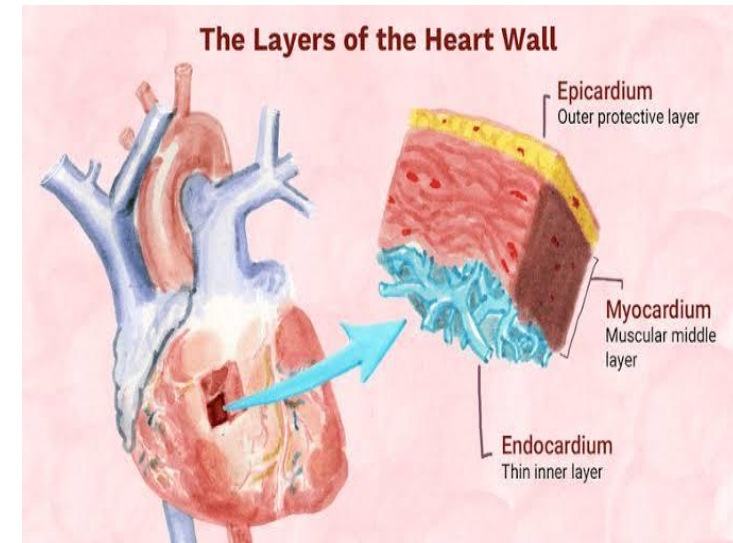
Second heart sound,
“dub”, occurs when
semilunar valves close



Semilunar
valves

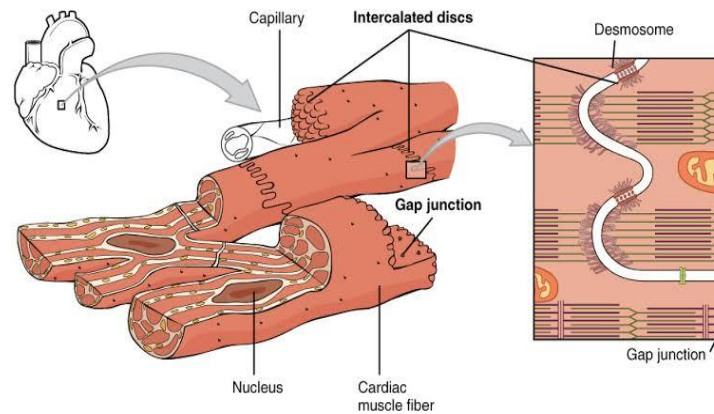
The wall of the heart has three layers:

1. **Epicardium** (The outermost layer).
3. **Myocardium** (The middle layer). (muscular layer)
2. **Endocardium** (The innermost layer).



How the heart contract?

- When the action potential reaches cells.
- *This stimulates the release of **Ca²⁺** from the *sarcoplasmic reticulum*.*
- **Ca²⁺** *promotes* the interaction of actin and myosin → contraction

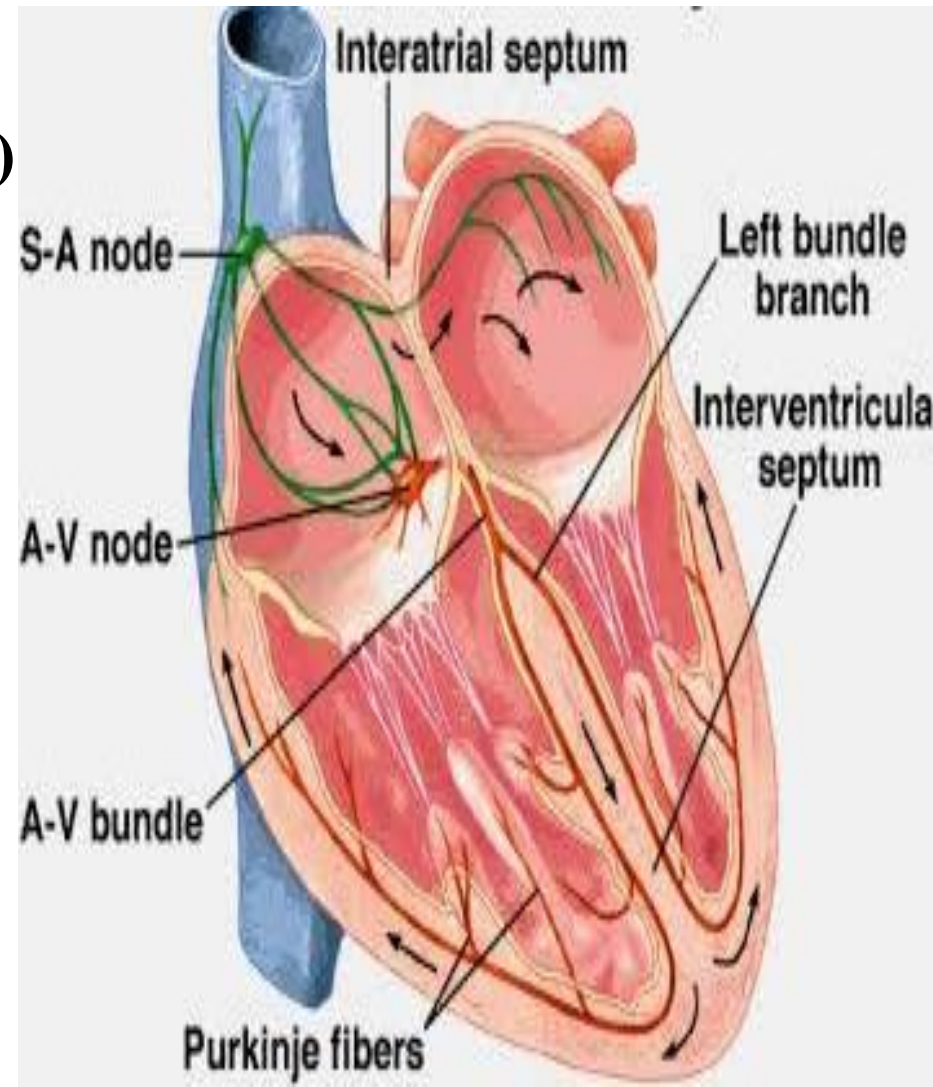


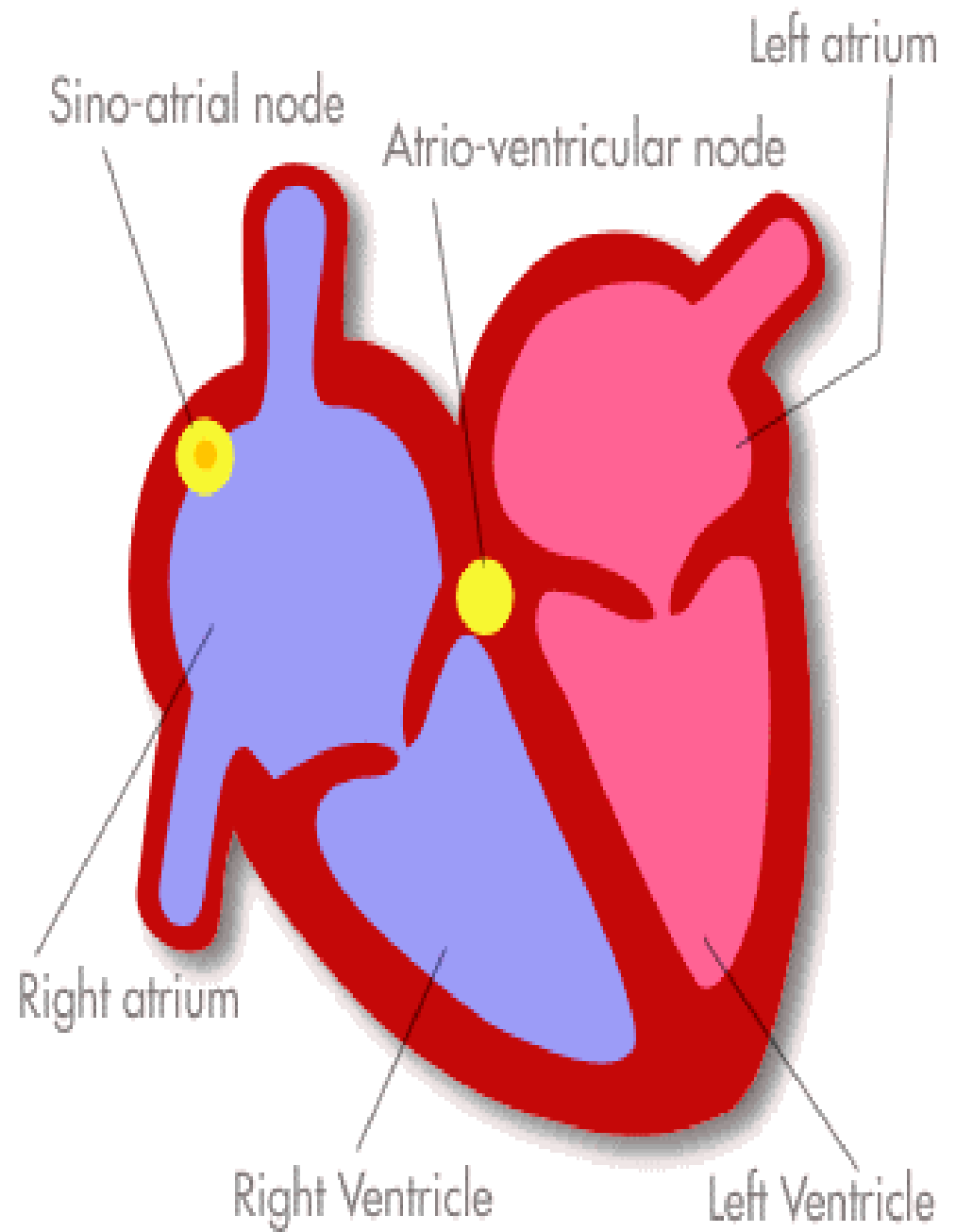
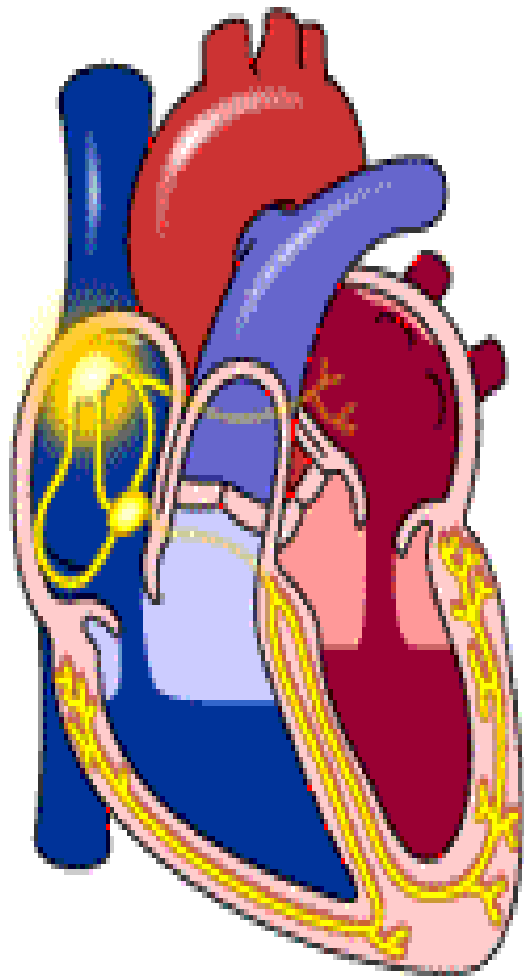
Electrical activity of the heart

Conducting system

The specialized excitation and electrical conduction system in the heart consists of the following:

- **Sinoatrial (SA) node (the pacemaker)**
- **Atrioventricular (AV) node**
- **AV Bundle (Bundle of His)**
- **Bundle branches**
- **Purkinje fibers**





Physiological properties of heart

1-Automaticity:

- It is the ability of the heart to generate electrical stimulus
- generated from sinoatrial node (SAN)

2-Rhythmicity:

It is the ability of the heart to beat regularly (72beat/min).

3-Contractility:

It is the ability of the heart to contract and push blood into the circulation.

4-Excitability:

It is the ability of the cardiac muscle to respond to electrical stimulus by contraction.

5-Conductivity:

It is the ability of the cardiac muscle to conduct electrical stimulus from one part of the heart to another by way of conducting system.



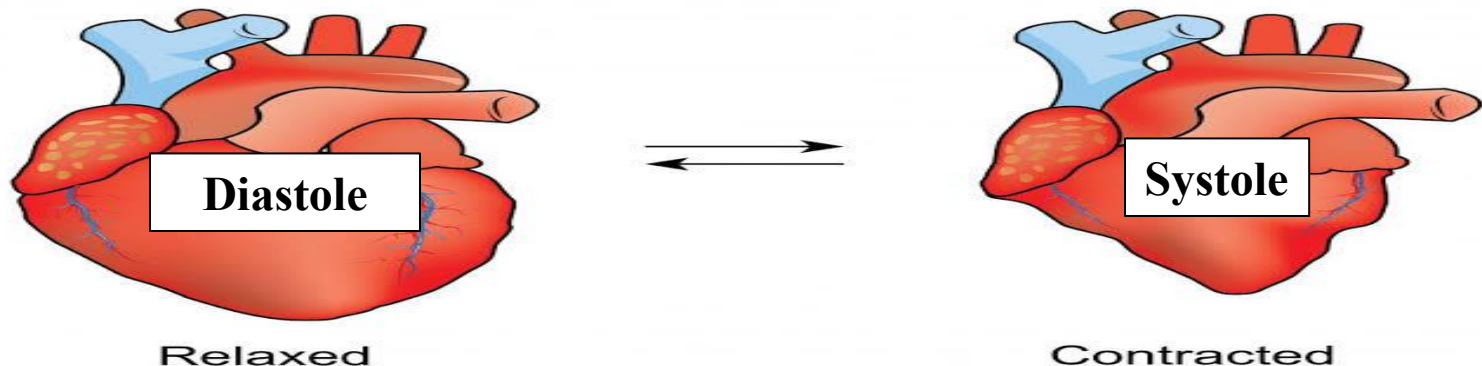
Cardiac cycle

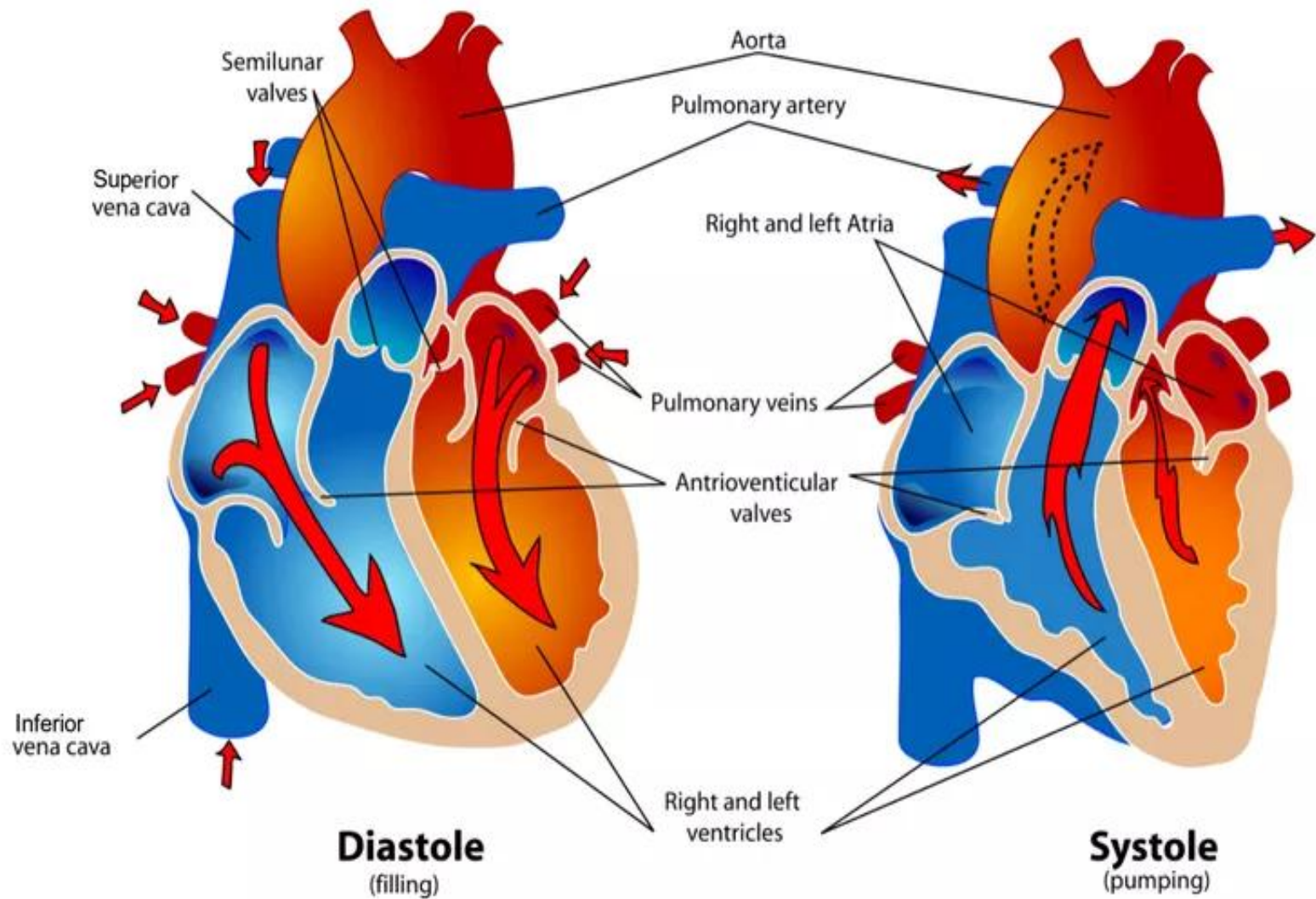
A-Diastole:

- 1- Blood collected in the **atria**
- 2- (**S-A node**) sends out an electrical signal that causes the atria to contract.
- 3- This contraction pushes blood into the resting.**ventricles**

B-Systole:

- 1- **Ventricles** are full of blood
- 2- The electrical signals from the S-A node travel along a pathway of cells to the ventricles through A-V node, bundle of His and Purkinje fibers
- 3- Causing them to **contract** and push the blood through the **pulmonary and aortic valves**.





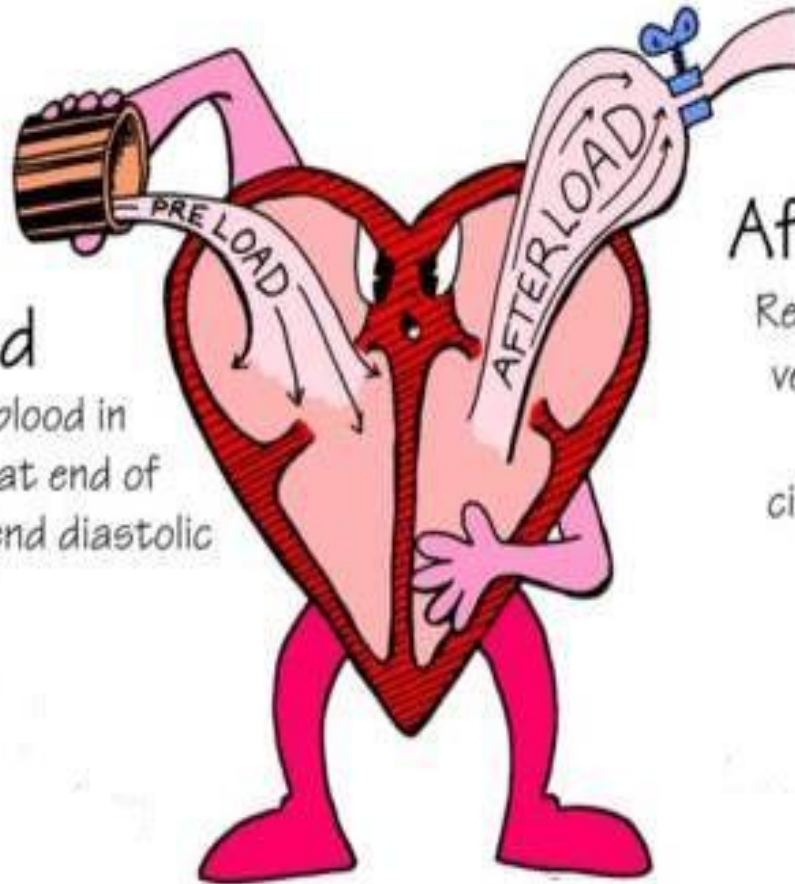
There are two loads on left ventricle during the cycle: preload and afterload

Preload	After load
The load that makes the ventricular muscles at its maximum length, Occurs at the end of the diastole	amount of resistance the heart must overcome to open the aortic valve and push the blood volume out into the systemic circulation
It depends on the volume of blood collected in the atrium (called venous return)	- It depends on the resistance of the arteries to blood pumping from ventricles (called peripheral resistance)

PRELOAD AND AFTERLOAD

Preload

Volume of blood in ventricles at end of diastole (end diastolic pressure)



Afterload

Resistance left ventricle must overcome to circulate blood